L3 Data Specialist

Module 6: From Data Citizen to Data Specialist

Three-day class-based learning

QA

Learn. To Change.



Schedule: Day 2

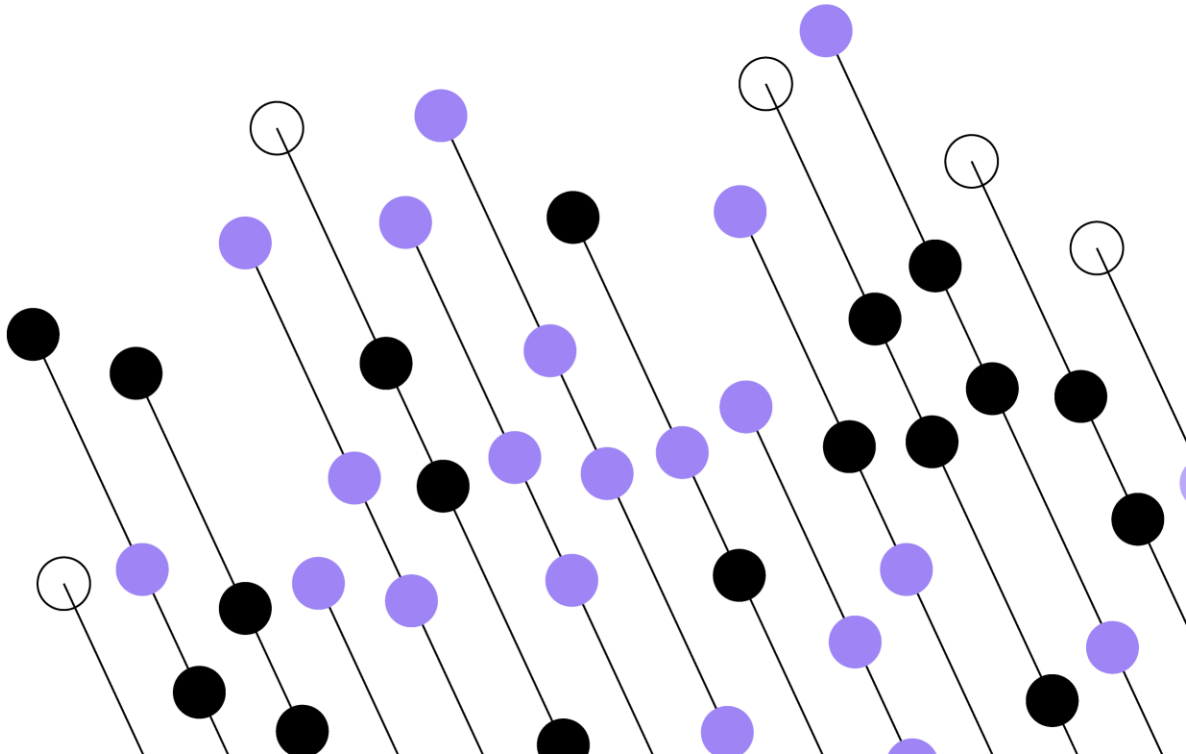
AM

- EPA methods and structure
- EPA prep: Scenario demonstration practice (part guided)

PM

- Continued: EPA prep: Scenario demonstration practice (part guided)

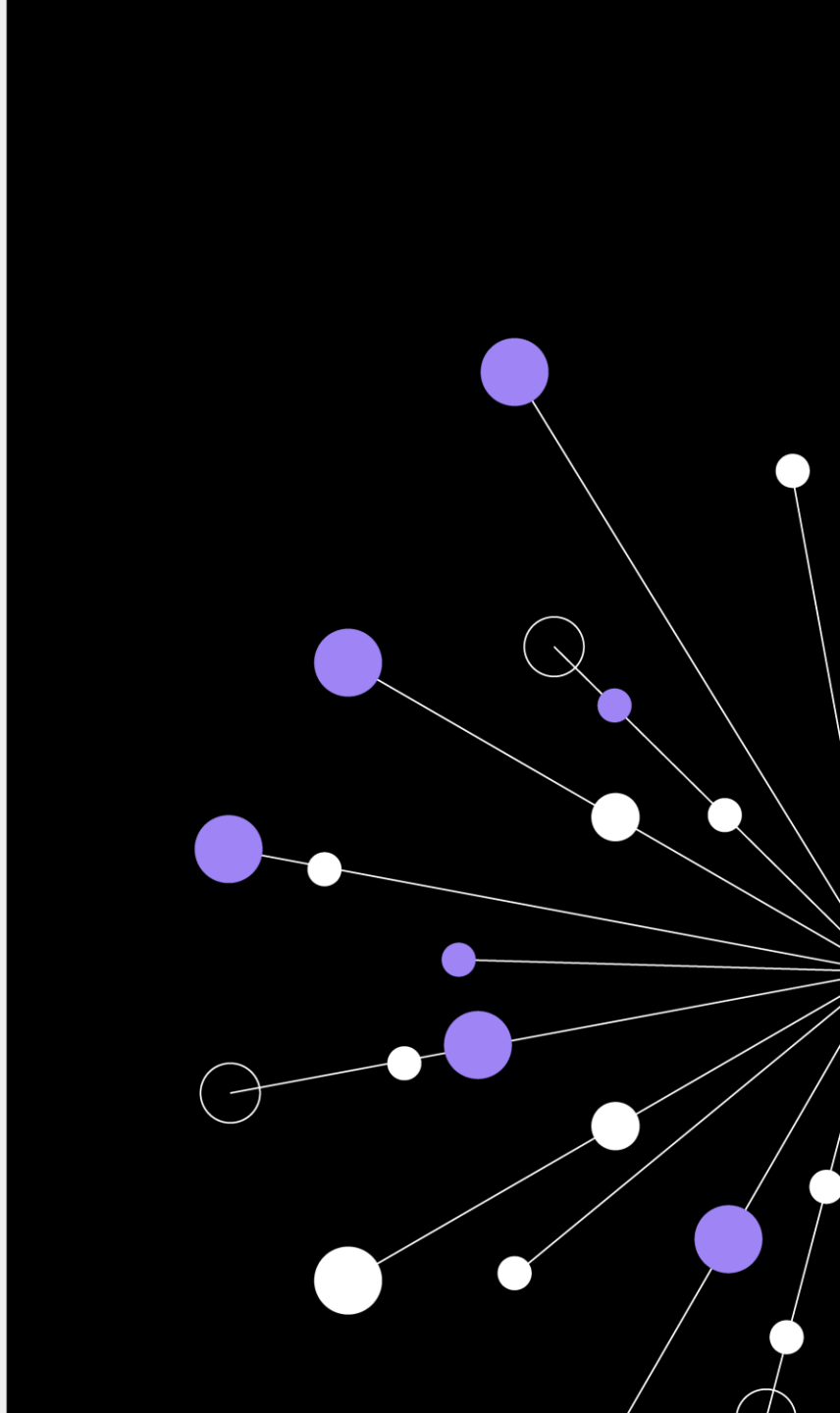
Day 2



End-Point Assessment (EPA)

This is an opportunity for us to discuss the EPA, evidence collection, and provide transparency.

- Types of EPA evidence
- Scenario demonstrations
- Review



End-Point Assessment (EPA)

This EPA has two assessment methods. The grades available for each assessment method are below.

- **Assessment method 1 — Scenario demonstration with questioning:** Fail / Pass / Distinction.
- **Assessment method 2 — Professional discussion underpinned by a portfolio of evidence:** Fail / Pass / Distinction.
- The result from each assessment method is combined to decide the **overall apprenticeship grade**. The following grades are available for the apprenticeship: Fail / Pass / Merit / Distinction.

EPA AM1: Scenario demonstration with questioning

On the day of the scenario demonstration, the independent assessor must explain to the apprentice the format and timescales of the scenario test and questions before it starts. The assessor will also provide the apprentice with an additional dataset related to the data set they were given to ensure a fair and consistent assessment.

Apprentices must be provided with both written and verbal instructions on the tasks they must complete. This does not count towards the assessment time.

The scenario demonstration with questioning will take 90 minutes. The EPAO must ensure that all scenarios provided meet the KSBs mapped to this method. The scenario may use datasets that are in a different business domain to the one in which the apprentice normally works.

There will be a minimum of one scenario which must enable the apprentice to demonstrate the following:

- Data gathering
- Data analysis
- Data validation

These activities provide the apprentice with the opportunity to demonstrate the KSBs mapped to this assessment method.

EPA AM1: Scenario demonstration with questioning

Typical scenarios:

Scenario demonstration 1: Data gathering

Example content:

- Access multiple datasets from different sources.
- Blend or correlate these datasets via common fields to generate an aggregated dataset. Store this aggregated dataset for later analysis.
- Review the aggregated dataset for obvious trends or patterns, documenting the findings for use by those conducting analysis on it — the value of data to the organisation.

Scenario demonstration 2: Data analysis and validation

Example content:

- Access a dataset and filter it to contain only those records stated as being relevant for the purpose.
- Examine the filtered dataset and document primary statistics about the records and their most important fields (as related to the stated data purpose).
- Cleanse the filtered dataset of errors, correcting those where appropriate (for example, removing duplicates, fixing spelling mistakes), highlighting others for further review (for example, outliers), documenting all actions taken
- Conduct field encoding and/or structural normalisation actions to improve efficiency of storage and/or analysis.

EPA AM1: Scenario demonstration with questioning

Theme KSBs	Pass: Apprentices must demonstrate all of the pass descriptors	Distinction: Apprentices must demonstrate all of the pass descriptors and all of the distinction descriptors
Data gathering **K4** **K5** **K6** **K13** **S1** **S2** **S3** **S4**	Explain different data element formats and how their selection can impact precision, analysis and communication. **(K4)** Demonstrate how to access and extract data as well as select and migrate data from identified sources. **(K5, S1)** Demonstrate how to collate and format data in line with organisational standards. Be able to collect, format, and save datasets whilst being able to summarise and explain their gathered data. **(K6, S2, S3)** Demonstrate how to combine data from multiple sources using tools to identify trends. Is able to present these in an appropriate format and enable manipulation of data sets as required. **(K13, S4)**	Demonstrate a comprehensive understanding of collating, formatting, and saving data in strict adherence to industry standards, while efficiently analysing, summarising, and explaining the gathered data in a highly organised, systematic, and precise manner. **(K6, S2, S3)**

EPA AM1: Scenario demonstration with questioning

Theme KSBs	Pass: Apprentices must demonstrate all of the pass descriptors	Distinction: Apprentices must demonstrate all of the pass descriptors and all of the distinction descriptors
Data analysis and validation **K14** **K15** **K16** **K26** **S5** **S6** **S7** **S16**	Demonstrates how to audit data results to ensure accuracy, completeness, consistency, and traceability from original data. Understands how data analysis and querying tools can answer questions, solve problems, and have the potential to use automation for repeated data manipulation. **(K14, S7)** Demonstrates how to filter details by focusing on information relevant to the data tasks and purpose, while identifying faults and cleansing the data. **(K15, S6)** Uses basic statistical methods to extract relevant information from both structured and unstructured data. **(K16, S5)** Demonstrates the use of tools or methods that can or have been applied as prompts to research and evaluate data transformation techniques. **(K26, S16)**	Identify how automation could be used for repeatable data manipulation to improve efficiency and accuracy. **(K14, S7)**

EPA AM2: Professional discussion underpinned by portfolio

The professional discussion must last for 60 minutes.

In the professional discussion, an independent assessor and apprentice have a formal two-way conversation. It gives the apprentice the opportunity to demonstrate the KSBs mapped to this assessment method.

The professional discussion must be structured to give the apprentice the opportunity to demonstrate the KSBs mapped to this assessment method to the highest available grade.

An independent assessor must conduct and assess the professional discussion.

The purpose of the independent assessor's questions will be to assess the apprentice's competence against the following themes:

- Data gathering.
- Data analysis.
- Data validation.
- Distribution and dissemination.
- Approaches to work.

EPA AM2: Professional discussion underpinned by portfolio

Overview

This assessment will take the form of a professional discussion which must be appropriately structured to draw out the best of the apprentice's competence and excellence. It will involve questions that focus on the KSBs mapped to this assessment method.

The rationale for this assessment method is:

A professional discussion is a two-way dialogue between the apprentice and independent assessor. It allows the apprentice to evidence how they have met the KSBs which are underpinned by evidence drawn from their portfolio. A professional discussion is a well-recognised method widely used within the digital sector. It allows for Knowledge, Skills and Behaviours that may not naturally occur as part of another assessment method to be assessed and discussed more easily. The apprentice can draw upon other supporting evidence in the portfolio and can effectively determine the authenticity of that supporting evidence.

As a structured, in-depth two-way conversation between the independent assessor and apprentice, the professional discussion will provide assessment of understanding the Knowledge, Skills and Behaviours.

EPA AM2: Professional discussion underpinned by portfolio

Theme KSBs	Pass: Apprentices must demonstrate all of the pass descriptors	Distinction: Apprentices must demonstrate all of the pass descriptors and all of the distinction descriptors
Data analysis and validation **K17** **K18** **S12**	Demonstrates testing and assessing the integrity of data against standard formats while explaining common data quality issues that can arise, methods for validating data to mitigate these issues, and corrective actions that can be taken. **(K17, K18, S12)**	N/A

EPA AM2: Professional discussion underpinned by portfolio

Theme KSBs	Pass: Apprentices must demonstrate all of the pass descriptors
Distribution and dissemination **K9** **K10** **K11** **K12** **K19** **K21** **K22** **S8** **S9** **S10**	Demonstrate management and presentation tools to visualise data, collaborate using communication technologies to share findings, and apply appropriate methods to help audiences understand data insights and their implications, tailoring communication to meet audience needs. **(K9, K10, K11, S8)** Identifies roles within an organisation that require access to data or an understanding of data findings and determine how these roles influence the level of detail needed in data communications. Be able to produce clear and consistent documentation of the data shared and actions taken, using standard organisational templates where applicable or required. **(K12, S9)** Outlines the value of data to an organisation. **(K21)** Explains how they store, manage, and distribute data in compliance with legal and regulatory requirements, ensuring adherence to organisational, national and sector standards, and applicable legislation. **(K19, S10)** Understands the significance of cultural awareness, diversity, and accessibility with respect to datasets and any bias that may be present in them. **(K22)**

EPA AM2: Professional discussion underpinned by portfolio

Theme KSBs	Distinction: Apprentices must demonstrate all of the pass descriptors and all of the distinction descriptors
Distribution and dissemination **K9** **K10** **K11** **K12** **K19** **K21** **K22** **S8** **S9** **S10**	Shows a deep understanding of the various roles within an organisation that require access to data or an understanding of data findings. Expertly determines how these roles impact the level of detail required in data communications, while consistently producing clear, comprehensive, and well-documented reports on the data shared and actions taken, adhering to standard organisational templates where applicable or required. **(K12, S9)** Evaluates advanced strategies for storing, managing, and distributing data in strict adherence to evolving data security standards, sustainable best practices, and comprehensive legislation, while proactively addressing emerging compliance challenges, **K19, S10**.

EPA AM2: Professional discussion underpinned by portfolio

Theme KSBs	Pass: Apprentices must demonstrate all of the pass descriptors	Distinction: Apprentices must demonstrate all of the pass descriptors and all of the distinction descriptors
Approaches to work **K24** **K25** **S11** **S13** **S14** **S15** **B1** **B2** **B3** **B4**	Demonstrates how they prioritise activities based on the duties to be performed, considering any known or expected impact on others. Working independently to meet deadlines and manage stakeholder expectation. Explains how they prioritise activities taking account of others when working towards a common goal or on the same datasets. **(S13, S14, B1, B2)** Explains how they follow policies and procedures related to environmental impact and sustainability, adhere to equity, diversity, and inclusion policies for a common goal, consider ways to reduce impact, support social inclusion in the workplace, and take responsibility for acting sustainably in their role. **(K24, K25, S11, S15, B3, B4)**	N/A

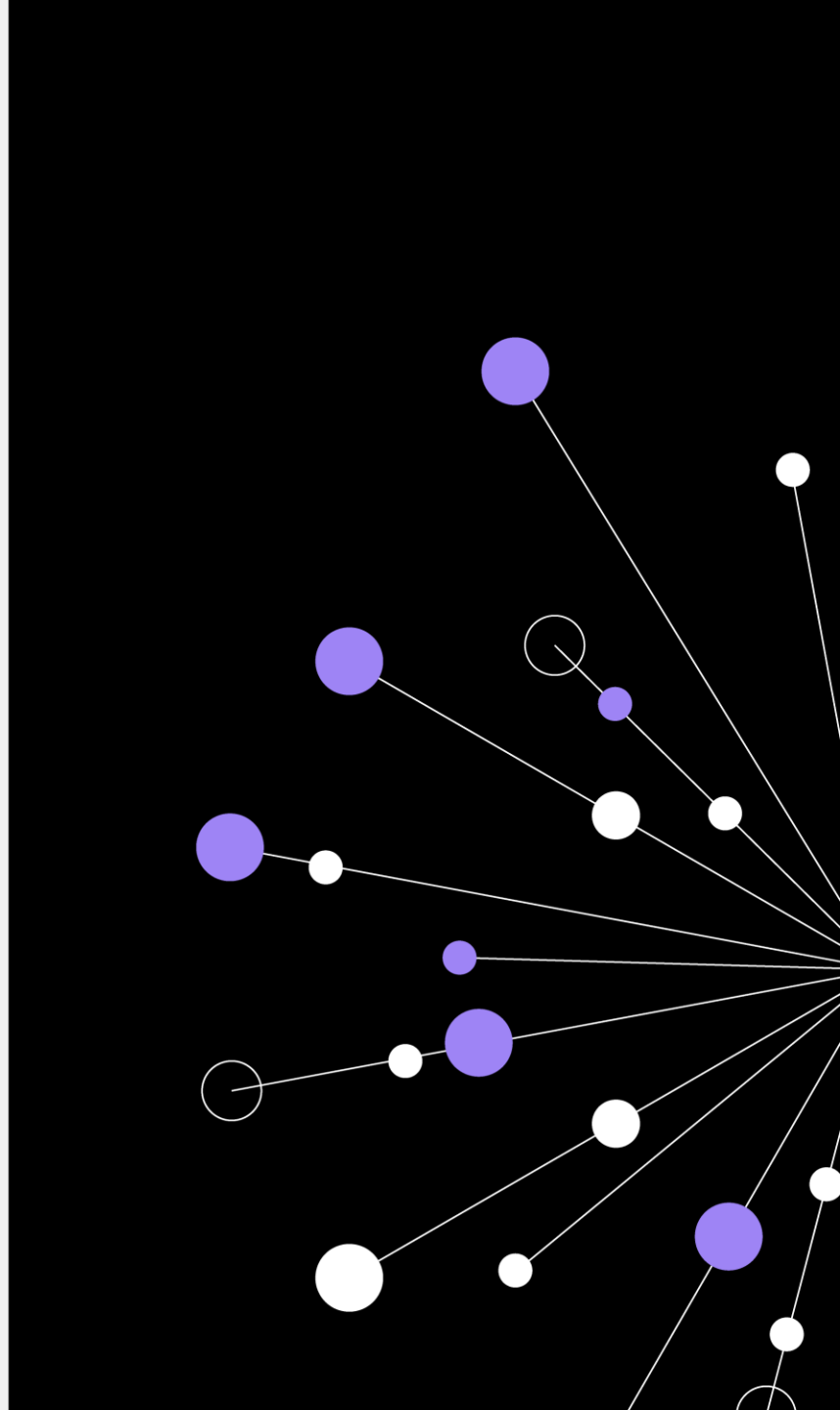
EPA AM2: Professional discussion underpinned by portfolio

Theme KSBs	Pass: Apprentices must demonstrate all of the pass descriptors	Distinction: Apprentices must demonstrate all of the pass descriptors and all of the distinction descriptors
Data gathering **K1** **K2** **K3** **K7** **K8** **K20** **K23**	Identifies types of data and common sources of data. **(K1, K2)** Identifies types of data storage formats and their importance for analysis. **(K3)** Explains the importance of anonymising data and how to do so. **(K7, K8)** Explain the relationships between data, machine learning, AI, and generative AI, and how ethical data use in AI and automated systems impacts organisations, including the potential consequences of unethical data use. **(K20, K23)**	N/A

Activity for EPA’s preparation

QA scenario demonstration practice

- The QA scenario demonstration practice is an Excel-based self-contained activity that aims to provide learners additional practice for their End-Point Assessment (EPA), particularly in relation to the scenario demonstration.
- The activity will be part guided and part independent.
- The activity was not designed to be completed within the time limits of the EPA, but it aims to support learners in practising more and further familiarising with tools and techniques which will be useful for their EPA.
- Learners may undertake the activities using any application or tool that they wish. However, Excel and Power Query are recommended.



EPA preparation: Part guided

In this activity, we will apply the KSBs assigned to this assessment method and apply them to a part guided, part independent activity. This is vital revision practice for your EPA assessment method 1.

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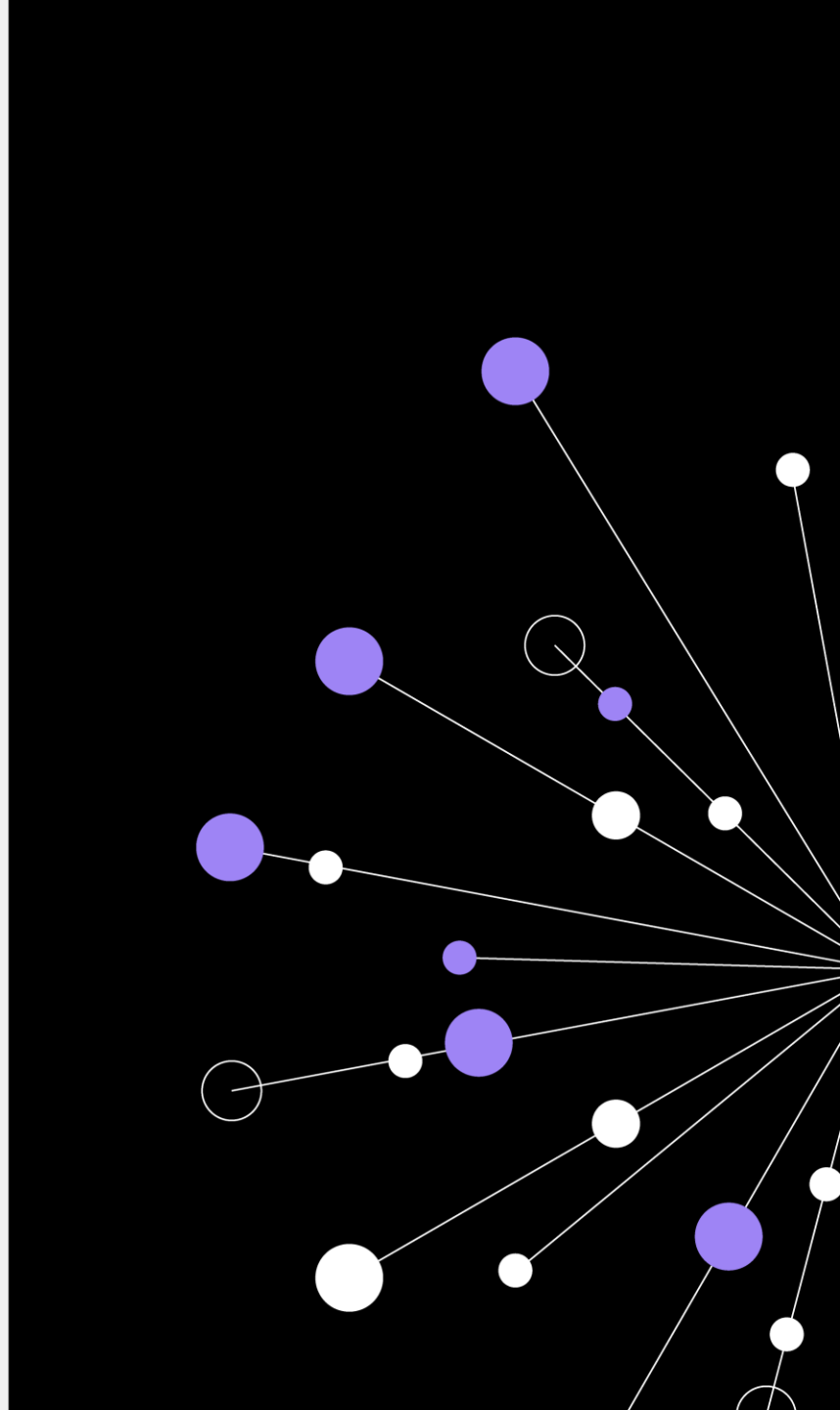
‘Day 2 EPA PREP - Scenario Demonstration Practice ACTIVITY.xlsx’

‘Day 2 DATA’

Optional:

‘Day 2 EPA Prep - Scenario Demonstration Practice - QUESTIONS.pdf’

‘Day 2 EPA Prep - Scenario Demonstration Practice GUIDELINES.pdf’



1:1 support



Up to 15 minutes of 1:1 support time has been integrated into the Day 2 workshops.



In preparation, please ensure you can demonstrate what online learning tasks you have engaged with/completed.